PHYS 616 Mathematical Methods for Physicists

Fall 2026

Instructor:	Cody Schimming (ISC 4120)	**Time:**	Tu/Th 7:00 – 9:00 PM
Email:	cody.schimming@umb.edu	**Place:**	University Hall Y04-4100

Office Hours: Held in ISC 4120, Thursdays 3:00 – 4:00 PM or by appointment.

Course Description: Mathematics is one of the most important tools in a physicist's arsenal. In this course, we will survey different aspects of mathematics and how they relate to the study of physics. The focus will be on conceptual understanding and problem solving, rather than proofs and derivations. We will also try to put mathematical concepts, results, and methods in the context of physical understanding. The goal is to expose you to mathematical notation, concepts, results, and methods so that you are able to use them in your own work and recognize them when reading physics literature.

Course Outline:

- Linear Algebra (Weeks 1–2)
- Tensor Calculus (Weeks 2–4)
- Functional Calculus (Weeks 4–5)
- Complex Analysis (Weeks 6–8)
- Differential Equations (Weeks 9–12)
- Various Topics* (Weeks 12–14)

The schedule of topics may change as the semester develops. *Topics might include Group Theory, Differential Geometry, Probability Theory, Algebraic Topology, Chaos Theory. The choice of which topics are covered will be guided by class interest.

Textbooks: There is no required textbook for this course. Some helpful references that will cover most of the material are

- G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists*
- S. Hassani, *Mathematical Methods for Students in Physics and Related Fields*
- N. A. Papadopoulos and F. Scheck, *Linear Algebra for Physics*

Lecture notes will also be provided and will include additional suggested reference material.

Homework: Apart from the first week and the week of Thanksgiving, there will be weekly homework assignments due by the end of the day each Friday and can be turned in in-person or by emailing me (please make sure that the scans or pictures are legible). The homework will include problems that will allow you to practice the methods we cover in class as well as provide more depth and opportunities to explore beyond what is covered in class. It is essential that you attempt to do the homework yourself. For this purpose, homework will be graded based on serious and honest participation, and marked homework with feedback will be returned.

Tests: There will be three tests, tentatively scheduled for the first Thursday of each month:

- Test 1: Oct. 1
- Test 2: Nov. 5
- Test 3: Dec. 3

There will not be a final exam. The tests will be structured as a few problems similar to those on the homework sets. Students will be given one hour out of the class time to complete the test. If you plan to be absent on a test day, please let me know in advance so we can schedule another time for you to take it.

Grading:

- Homework: 40%
- Tests: 60% (20% each)

Course Grade Percentages (may be adjusted):

A: 90% A-: 85% B+: 80% B: 75% B-: 70% C+: 65% C: 60% C-: 55%

AI Policy: Students should not use generative AI tools such as chatGPT to fully complete homework problems. Students may use generative AI tools as learning aids or to clarify concepts, though any work that is turned in is the student's responsibility. If generative AI is used, the student must note where and how it was used on an assignment. If it is suspected that a student used generative AI to fully complete homework problems, that student may forfeit credit and feedback for that assignment.

Accommodations: UMass Boston is committed to providing reasonable academic accommodations for all students with disabilities. Please contact the Ross Center for Disability Services (ross.center@umb.edu, 617-287- 7430) for recommendations for specific accommodations if needed. If you already have a recommendation, please contact me as soon as possible, preferably within the first two weeks of classes, so that we can work out the best way to support you.

Studen Mental Health: Graduate school can be a significant undertaking. If you or someone you know experiences academic stress, difficult life events, or feelings of anxiety or depression, we encourage you to seek support. Helpful, effective resources are available via the University Health Services. The UHS Counseling Center can be reached at 617-287-5690 (including after-hours and on weekends). Whether or not you are a current patient at the center, you will be able to access telehealth crisis support. Please know that the Counseling Center and university are working to provide remote support for students during this difficult time. More information is available online at https://www.umb.edu/healthservices/counseling_center.

Academic Integrity: Education at UMass Boston is sustained by academic integrity. Academic integrity requires that all members of the campus community are honest, trustworthy, responsible, respectful, and fair in academic work at the university. As part of being educated here, students learn, exercise, increase, and uphold academic integrity. Academic integrity is essential within all classrooms, in the many spaces where academic work is carried out by all members of the UMass Boston community, and in our local and global communities where the value of this education fulfills its role as a public good. Students are expected to adhere to the Student Code of Conduct, including policies about academic integrity, delineated here: https://www.umb.edu/life_on_campus/dean_of_students/student_conduct.